

Prompt Documentation — Zulqarnain Academy Website Development

Project Overview

I developed a complete educational platform named **Zulqarnain Academy** using modern web technologies. The website was designed to connect students, tutors, consultants, and customer service agents through an interactive and professional online system.

The project includes:

- Frontend Website
- Backend API System
- MongoDB Database Integration
- Authentication System
- OTP Password Reset System
- File Upload System
- Responsive Mobile Design
- Professional UI/UX
- Career Management System
- Course Enrollment System

Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript (Vanilla JS)

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas
- MongoDB Compass
- Mongoose ODM

Additional Packages

- Multer (File Upload)
- Nodemailer (OTP Email System)
- CORS
- dotenv
- bcrypt

Main Development Prompts Used During Website Creation

1. Homepage Design Prompt

Create a modern educational website homepage named “Zulqarnain Academy” with a professional navbar, hero section, services section, tutor cards, animated buttons, responsive layout, footer, and modern glassmorphism UI using HTML, CSS, and JavaScript. The website must look premium, mobile responsive, and visually attractive for students and tutors.

2. Responsive Navbar Prompt

Create a fully responsive navigation bar with a mobile hamburger menu. The navbar should collapse properly on mobile devices and open smoothly when clicking the three-line menu icon. Include dropdown functionality for career pages.

3. Become a Tutor Page Prompt

Design a professional “Become a Tutor” page containing:

- Tutor application form
- Qualification field
- Subject field
- Experience selection
- Teaching mode selection
- City field
- Professional introduction textarea
- CV upload option
- Commission policy popup modal
- Responsive mobile layout
- Backend connection using Node.js and MongoDB
- File upload functionality using Multer

4. Tutor Backend API Prompt

Create a backend API route using Express.js and MongoDB to store tutor application data securely. Include:

- File upload system using Multer
- Form validation

- MongoDB schema
- API route /api/tutor/apply
- Error handling
- CV storage in uploads folder
- Database insertion functionality

5. Course Enrollment System Prompt

Create a course enrollment page with:

- Student enrollment form
- Course selection dropdown
- Email validation
- MongoDB storage
- Backend API integration
- Responsive design
- Professional layout
- Student data saving functionality

6. Consultant Career Page Prompt

Create a professional Educational Consultant career page with:

- Career details section
- Professional application form
- Multiple file uploads
- Experience field
- CV upload
- Backend integration with MongoDB
- Fully responsive design
- Animated UI

7. Customer Service Agent Career Prompt

Create a Customer Service Agent application page including:

- Professional job description cards
- Application form
- Education field
- Experience certificate upload
- Communication experience textarea
- Backend API connection
- MongoDB data storage
- Responsive design

8. Forgot Password OTP System Prompt

Create a complete OTP-based forgot password system using:

- Nodemailer
- Gmail SMTP
- OTP generation
- OTP verification
- Password reset system
- MongoDB user schema update
- Frontend popup modal
- Backend API routes
- Secure password hashing using bcrypt

9. MongoDB Integration Prompt

Connect the Zulqarnain Academy backend with MongoDB Atlas and MongoDB Compass using Mongoose. Create schemas and models for:

- Users
- Tutors
- Consultants
- Customer Service Agents
- Course Enrollments

Ensure all data stores successfully in the database.

10. Authentication System Prompt

Create a login and registration system using Node.js, Express.js, MongoDB, and bcrypt password hashing. Include:

- User registration
- Secure login
- Email uniqueness validation
- Password encryption
- Error handling
- Session/token preparation

11. Mobile Responsiveness Prompt

Make the entire website fully responsive for mobile devices. Remove horizontal scrolling, fix overflowing sections, optimize navbar behavior, resize images properly, and improve mobile user experience using CSS media queries.

12. Performance Optimization Prompt

Optimize website performance by:

- Compressing large images
- Converting PNG to jpeg
- Lazy loading images
- Improving page speed
- Reducing mobile lag
- Optimizing CSS layout

13. Error Debugging Prompt

Analyze and fix backend and frontend connection issues including:

- 404 API errors
- Cannot POST errors
- MongoDB connection problems
- Express route issues
- JSON parsing errors
- Server crashes
- Undefined variables
- File upload failures

14. Final Year Project Documentation Prompt

Create professional final year project documentation for Zulqarnain Academy including:

- Abstract
- Introduction

- Objectives
- Functional requirements
- Non-functional requirements
- ER diagram
- UML diagrams
- Database design
- Testing
- Future improvements
- Conclusion

15. ER Diagram & UML Prompt

Generate ER diagrams and UML diagrams for Zulqarnain Academy educational platform showing:

- Users
- Tutors
- Courses
- Enrollment
- Consultant Applications
- Customer Service Agents
- Authentication system
- Database relationships

Development Process Followed

Step 1 — Frontend Development

Created all website pages using HTML, CSS, and JavaScript.

Step 2 — Responsive Design

Implemented media queries and mobile responsiveness.

Step 3 — Backend Setup

Configured Node.js and Express.js server.

Step 4 — Database Integration

Connected MongoDB Atlas and MongoDB Compass.

Step 5 — API Development

Created REST APIs for forms and authentication.

Step 6 — File Upload System

Implemented Multer for CV and certificate uploads.

Step 7 — Authentication

Developed login, registration, and OTP password reset systems.

Step 8 — Debugging & Testing

Fixed routing errors, backend crashes, and API connection problems.

Step 9 — Optimization

Improved performance, responsiveness, and UI consistency.

Features of Zulqarnain Academy

- Tutor Hiring System
- Student Enrollment System
- Consultant Recruitment
- Customer Service Hiring
- Authentication System
- OTP Password Recovery
- File Upload System
- MongoDB Cloud Database
- Responsive Design
- Professional User Interface
- Mobile Optimization

Learning Outcomes

Through this project I learned:

- Full Stack Web Development
- REST API Development
- MongoDB Database Design
- Frontend & Backend Integration
- Authentication Systems
- Error Handling
- Responsive Web Design
- File Upload Management
- Real-world Project Deployment Concepts

Conclusion

The Zulqarnain Academy project demonstrates a complete full-stack educational management platform developed using modern web technologies. The project combines frontend design, backend development, database management, authentication systems, and responsive UI/UX principles to create a professional and scalable educational platform suitable for real-world implementation.